The mixed remainder with arbitrary shifts

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Abstract

The mixed term of the anchored decomposition (unit 88) was bounded by the constant-kernel block with both exponents shifted by one. Here the bound is generalised to any amplitude dominated by $Cu^{M_1}v^{M_2}(1+s)^D e^{\beta sL}$ on the box: its block integral is at most $Ce^{D^2/\beta} \cdot \text{twoDGeneral}(\beta/2, 2L, b, N^2, h_1+M_1, h_2+M_2, k_1, k_2)$, which is $O\big(N^{-\min((h_1+M_1+1)/k_1,(h_2+M_2+1)/k_2)}(1+\log N)\big)$. With $M_1/k_1, M_2/k_2 > \tau$ this is the remainder estimate $O(N^{-(p+\tau)}(1+\log N))$ of the rectangular face-jet decomposition of the $d=2$ block. The polynomial envelope is absorbed by $(1+s)^D \le e^{D^2/\beta}e^{\beta s^2/2}$. Zero `sorry`/`axiom`.

1 The things formalised

```
theorem one_add_pow_le_exp (  s :  ) (D :  ) (h  : 0 <  ) (hs : 0   s) :
    (1 + s) ^ D   Real.exp ((D :  ) ^ 2 /  ) * Real.exp (  / 2 * s ^ 2)
theorem twoDAmp_env_le (  b L C N :  ) (h  h  k  k  M  M  D :  ) (Φ :   →   →   →  )
    (h  : 0 <  ) (hN : 0   N) (hC : 0   C)
    (hΦ : Continuous fun x :   ×   ×   => Φ x.1 x.2.1 x.2.2)
    (hbound :   u   Icc (0 :  ) b,   v   Icc (0 :  ) b,   s, 0   s →
      |Φ u v s|   C * (u ^ M  * v ^ M ) * ((1 + s) ^ D * Real.exp (  * s * L))) :
    |twoDAmp   b N h  h  k  k  Φ|
       C * Real.exp ((D :  ) ^ 2 /  )
        * twoDGeneral (  / 2) (2 * L) b (N ^ 2) (h  + M ) (h  + M ) k  k
theorem twoDGeneral_shift_isBigO (  a b :  ) (h  h  k  k  :  ) (h  : 0 <  ) (hb : 0 < b)
    (hk  : 0 < k ) (hk  : 0 < k ) :
    (fun N :   => twoDGeneral   a b (N ^ 2) h  h  k  k ) =O[atTop]
      fun N :   => N ^ (-min (((h  :  ) + 1) / k ) (((h  :  ) + 1) / k )) * (1 + Real.log N)
theorem twoDAmp_env_isBigO ... :
    (fun N :   => twoDAmp   b N h  h  k  k  Φ) =O[atTop]
      fun N :   => N ^ (-min ((((h  + M  :  ) :  ) + 1) / k )
          ((((h  + M  :  ) :  ) + 1) / k )) * (1 + Real.log N)
```

2 Mathematics

Pointwise on the box, $u^{h_1}v^{h_2}e^{-\beta s^2}|\Phi| \le Cu^{h_1+M_1}v^{h_2+M_2}e^{-\beta s^2}(1+s)^D e^{\beta sL}$ with $s = Nu^{k_1}v^{k_2}$, and $(1+s)^D \le e^{Ds} \le e^{D^2/\beta}e^{\beta s^2/2}$ (the second step is $Ds \le D^2/\beta + \beta s^2/2$, i.e. $(\beta s/2 - D)^2 \ge 0$). The right side is then $Ce^{D^2/\beta}$ times the integrand of the constant-kernel block twoDGeneral with parameters $(\beta/2, 2L, N^2)$ and exponents (h_1+M_1, h_2+M_2), whose $N \to \infty$ behaviour is known from the unconditional trichotomy of unit 47 (`twoDGeneral_isBigO_min`) after composing with $N \mapsto N^2$ and converting $\log N^2 = 2\log N$.

3 Paper-facing meaning

In the rectangular decomposition $\Phi = \sum_{i<M_1} u^i a_i + \sum_{j<M_2} v^j b_j - \sum c_{ij} u^i v^j + R$ with $|R| \leq Hu^{M_1} v^{M_2}$ (Astra #4(b)), the remainder contributes $O(N^{-\min(p+M_1/k_1, p+M_2/k_2)}(1+\log N))$; choosing $M_1/k_1, M_2/k_2 > \tau$ makes it $O(N^{-(p+\tau)}(1+\log N))$, the remainder order of the τ-cutoff theorem. Together with the face-term expansions of unit 104 this is all the analysis needed; what remains is the decomposition identity and the bookkeeping.

4 Lean notes

- `field_simp` can close a goal outright; a trailing `ring` then errors with "no goals".
- After `Real.log_pow` the factor `↑2` is a `Nat` cast, invisible to `nlinarith`; `push_cast` first.
- `IsBigO.comp_tendsto` with `tendsto_pow_atTop two_ne_zero` transports an `atTop` estimate in n to one in $N = \sqrt{n}$; `Real.rpow_natCast` and `Real.rpow_mul` collapse $(N^2)^{-m/2}$ to N^{-m}.